

Roll No.:

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B.Tech

Semester: III

Name of the Course: Introduction to AI

Course Code: TCS-542

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Define Artificial Intelligence (AI). Discuss its evolution, key goals, and how it differs from traditional computing.	CO1
(b)	Design a basic AI system architecture for a personal assistant (like Alexa or Google Assistant). Explain the roles of perception, reasoning, knowledge base, and learning in your design.	CO2 CO1
(c)	Discuss the Applications of AI in at least three domains. For healthcare domain, describe a specific AI-based application and how it improves performance over human methods.	
Q2		
(a)	Explain the Water Jug Problem as an example of a state-space search problem in Artificial Intelligence. Illustrate the steps to reach the goal using production rules with a suitable example.	CO2 CO3
(b)	You are designing a pathfinding system for a delivery robot. Suggest how you would represent the environment as a state space and which search algorithm (BFS, DFS, A*, or Hill Climbing) would be most efficient. Justify your choice.	CO3
(c)	Compare Local Search Algorithms such as Hill Climbing, Simulated Annealing, and Local Beam Search. Also explain how each tries to overcome the problem of local maxima?	
Q3		
(a)	(a) Compare propositional and predicate logic. (b) Consider the propositions: - P: "Ram is intelligent." - Q: "Ram studies regularly." - R: "Ram scores high marks." Formulate logical expressions for the following and determine their logical equivalence: i) If Ram is intelligent and studies regularly, then he scores high marks.	CO4 CO4 CO4

	ii) If Ram does not study, then he will not score high marks.	
(b)	(a) Define uncertainty in AI. Why uncertainty is unavoidable in real world problems? (b) Discuss Full Joint Probability Distribution in detail.	
(c)	Knowledge Base (KB): 1. <i>Rain</i> $\rightarrow$ <i>WetGround</i> 2. <i>WetGround</i> $\rightarrow$ <i>SlipperyRoad</i> 3. <i>SlipperyRoad</i> $\rightarrow$ <i>AccidentProne</i> 4. Known fact: <i>Rain</i> (a) Using Backward Chaining, show the reasoning steps to determine whether <i>AccidentProne</i> can be inferred as true from the given facts. (b) Using Forward Chaining derive all the possible conclusions from the given knowledge base.	
Q4		
(a)	Represent the following English statements using Predicate Logic: 1. Socrates is a human. 2. Some teachers are researchers. 3. If a person is hardworking, then they will succeed. 4. Everyone hates someone. 5. If a person is humble and honest, he is respected.	CO5 CO5 CO5
(b)	Explain how Prolog represents knowledge? Also discuss the key symbols used in PROLOG along with their meaning.	
(c)	What are Semantic Networks and Partitioned Nets discuss in detail with the help of an example.	
Q5		
(a)	Discuss expert systems (MYCIN or DENDRAL) and explain: - What problem it was designed to solve - How it represented and processed knowledge	CO6 CO6 CO6
(b)	Explain Minsky's Frame Theory with an example of representing a "Restaurant Scene." Identify how subframes and slots help in structuring knowledge hierarchically.	
(c)	Explain the different types of meta-knowledge and discuss how it contributes to the self-improving capability of an expert system.	